Analysis report for sample: 13c7db25f08df3ef2a253e5e17b6f5fe

This report has been generated by a set of tools for semi-supervised behavioral analysis of executables.

More information about these reports at **http://www.vallejo.cc**.

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Report classification: 1642 (indicates the level of interest of the report, based on the amount of information collected about the sample).

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Sample

- md5: 13c7db25f08df3ef2a253e5e17b6f5fe

AV detections

- Microsoft: VirTool:Win32/CeeInject
- Kaspersky: Trojan-Downloader.Win32.Agent.hfvv
- Symantec: Trojan.Fareit!gen1
- Malwarebytes: Trojan.Kelihos.ED

Original source url

-

Virustotal url

- https://virustotal.com/es/file/5f54b4fe8b200a3fd3c390627a01200ecfd1e0c38c2a28ff3266422a9b9ba426/analysis

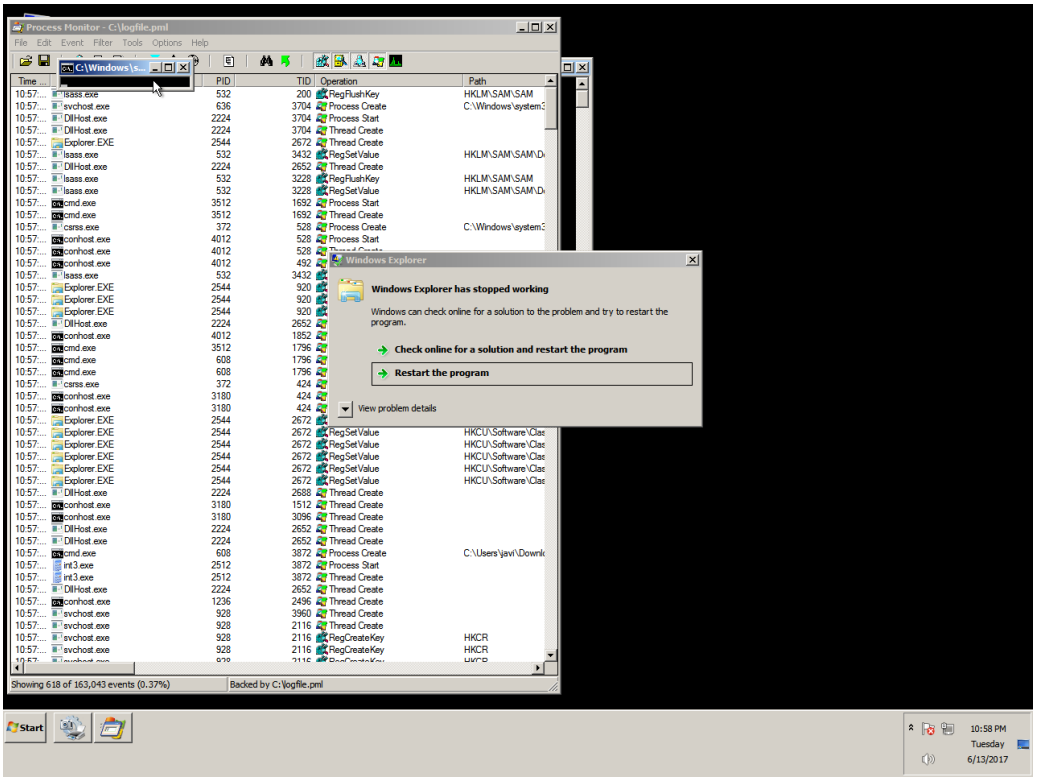
Yara matches

The following yara rules have matched injected or unpacked modules's code or data areas.

- Microsoft_Visual_Cpp_v50v60_MFC ()
- IsPE32 ()
- IsWindowsGUI ()
- HasOverlay (Overlay Check)
- ImportTableIsBad (ImportTable Check)
- HasRichSignature (Rich Signature Check)
- without_attachments (Rule to detect the no presence of any attachment)
- without_images (Rule to detect the no presence of any image)
- without_urls (Rule to detect the no presence of any url)
- contentis_base64 (This rule finds for base64 strings)
- suspicious_packer_section (The packer/protector section names/keywords)
- UPX ()
- Str_Win32_Winsock2_Library (Match Winsock 2 API library declaration)
- Str_Win32_Wininet_Library (Match Windows Inet API library declaration)
- Str_Win32_Internet_API (Match Windows Inet API call)
- PackerUPX_CompresorGratuito_wwwupxsourceforgenet ()
- UPX_wwwupxsourceforgenet_additional ()
- yodas_Protector_v1033_dllocx_Ashkbiz_Danehkar_h ()
- Netopsystems_FEAD_Optimizer_1 ()
- UPX_290_LZMA ()
- UPX_290_LZMA_Markus_Oberhumer_Laszlo_Molnar_John_Reiser ()
- UPX_290_LZMA_additional ()
- UPX_wwwupxsourceforgenet ()
- UPXv20MarkusLaszloReiser ()
- UPXV200V290MarkusOberhumerLaszloMolnarJohnReiser ()
- UPX20030XMarkusOberhumerLaszloMolnarJohnReiser ()
- UPX290LZMAMarkusOberhumerLaszloMolnarJohnReiser ()
- upx_3 (UPX 3.X)
- inject_thread (Code injection with CreateRemoteThread in a remote process)
- network_tcp_listen (Listen for incoming communication)
- network_tcp_socket (Communications over RAW socket)
- network_dns (Communications use DNS)

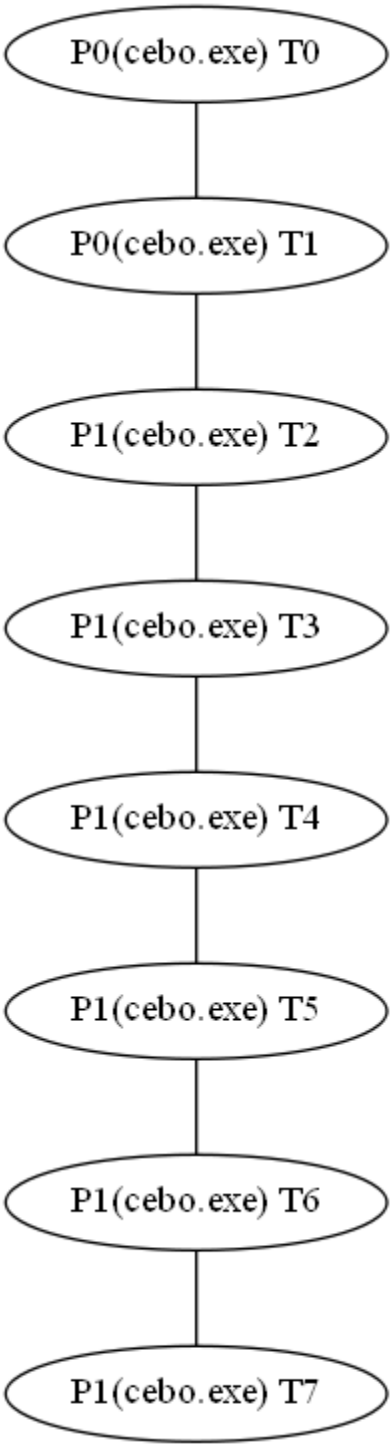
- win_files_operation (Affect private profile)

Snapshot


Process Monitor - C:\logfile.pml
File Edit Event Filter Tools Options Help
Windows Explorer
Windows Explorer has stopped working
Windows can check online for a solution to the problem and try to restart the program.
Check online for a solution and restart the program
Restart the program
View problem details
Showing 618 of 163,043 events (0.37%)
Backed by C:\logfile.pml
Start
10:58 PM
Tuesday
6/13/2017

Threads tree

The following tree represents sample's threads. T<index> is an alias for sample's threads (numeration is done in the order of threads creation). P<index> is an alias for processes owning sample's threads.



Running processes

Here it is the list of running processes after sample execution.

Processes with sample' threads

- Name cebo.exe Alias P0
- Name cebo.exe Alias P1

Other processes

- Name System
- Name smss.exe
- Name csrss.exe
- Name wininit.exe
- Name csrss.exe
- Name winlogon.exe
- Name services.exe
- Name lsass.exe
- Name lsm.exe
- Name svchost.exe
- Name svchost.exe
- Name svchost.exe
- Name svchost.exe
- Name svchost.exe
- Name audiodg.exe
- Name TrustedInstaller.exe
- Name svchost.exe
- Name svchost.exe
- Name spoolsv.exe
- Name svchost.exe
- Name svchost.exe
- Name vmtoolse.exe
- Name svchost.exe
- Name dllhost.exe
- Name msdtc.exe
- Name sppsvc.exe
- Name taskhost.exe
- Name dwm.exe
- Name explorer.exe
- Name vmtoolse.exe
- Name SearchIndexer.exe
- Name svchost.exe
- Name wmpnetwk.exe
- Name WmiPrvSE.exe
- Name chrome.exe
- Name WmiPrvSE.exe
- Name SearchFilterHost.exe
- Name taskhost.exe
- Name SearchProtocolHost.exe
- Name cmd.exe
- Name conhost.exe
- Name UI0Detect.exe
- Name UI0Detect.exe
- Name pxxx.exe
- Name dllhost.exe

- Name cmd.exe
- Name conhost.exe
- Name cmd.exe
- Name conhost.exe
- Name int3.exe
- Name explorer.exe
- Name explorer.exe
- Name svchost.exe
- Name WerFault.exe
- Name cmd.exe
- Name conhost.exe
- Name UI0Detect.exe
- Name cmd.exe
- Name conhost.exe
- Name cmd.exe
- Name conhost.exe
- Name explorer.exe
- Name explorer.exe
- Name cmd.exe
- Name conhost.exe
- Name taskkill.exe
- Name mode.com
- Name conhost.exe
- Name WmiPrvSE.exe
- Name cmd.exe
- Name conhost.exe
- Name cmd.exe
- Name conhost.exe
- Name UI0Detect.exe
- Name int3.exe

Most interesting actions

The following list it's a collection of the most interesting events captured. This list is ordered by the score assigned to the event. In the section "Threads behavioural information" it's possible to find all the actions performed by each sample's thread ordered chronologically.

- Process Create (C:\\Users\\javi\\Downloads\\cebo.exe PID: P1, Command line: "C:\\Users\\javi\\Downloads\\cebo.exe")
- RegCreateKey (HKCU\\Software\\Local AppWizard-Generated Applications Desired Access: Read/Write, Disposition: REG_CREATED_NEW_KEY)
- RegCreateKey (HKCU\\Software\\Local AppWizard-Generated Applications\\cebo Desired Access: Read/Write, Disposition: REG_CREATED_NEW_KEY)
- RegCreateKey (HKCU\\Software\\Local AppWizard-Generated Applications\\cebo\\Recent File List Desired Access: Read/Write, Disposition: REG_CREATED_NEW_KEY)
- RegCreateKey (HKCU\\Software\\Local AppWizard-Generated Applications Desired Access: Read/Write, Disposition: REG_OPENED_EXISTING_KEY)
- RegCreateKey (HKCU\\Software\\Local AppWizard-Generated Applications\\cebo Desired Access: Read/Write, Disposition: REG_OPENED_EXISTING_KEY)
- RegCreateKey (HKCU\\Software\\Local AppWizard-Generated Applications\\cebo\\Recent File List Desired Access: Read/Write, Disposition: REG_OPENED_EXISTING_KEY)
- RegCreateKey (HKCU\\Software\\Local AppWizard-Generated Applications\\cebo\\Settings Desired Access: Read/Write, Disposition: REG_CREATED_NEW_KEY)

Most interesting strings

The following list it's a collection of the most interesting strings found in the sample's modules code or data. A complete list of strings for each sample's module can be found in the section "Sample's unpacked / injected modules."

- url | valid domains | tlds: .ms-*nl
- url | valid domains | tlds: 7@.Tw
- url | valid domains | tlds: 3.vA#-v
- url | valid domains | tlds: 3.v\\9.vA
- api | frequently used by malware: GetTimeZoneInformation
- api | frequently used by malware: GetTickCount
- api | frequently used by malware: GetNetworkParams
- api | frequently used by malware: ExitProcess
- api | frequently used by malware: GetProcAddress
- api | frequently used by malware: Send
- api | frequently used by malware: CreateThread
- api | frequently used by malware: WriteFile
- api | frequently used by malware: OpenProcess
- api | frequently used by malware: Sleep
- api | frequently used by malware: recv
- api | frequently used by malware: send
- api | frequently used by malware: WSAStartup
- api | frequently used by malware: connect
- api | frequently used by malware: htons
- api | frequently used by malware: gethostbyname
- api | frequently used by malware: closesocket
- api | frequently used by malware: socket
- api | frequently used by malware: CreateRemoteThread
- api | frequently used by malware: EnumProcesses
- api | frequently used by malware: WriteProcessMemory
- api | frequently used by malware: ReadProcessMemory
- api | frequently used by malware: VirtualAllocEx
- api | wininet: InternetOpenA
- api | ws2_32: __WSAFDIsSet
- api | ws2_32: select
- api | ws2_32: WSASocketA
- api | ws2_32: WSAGetLastError
- api | ws2_32: ioctlsocket
- api | ws2_32: WSADuplicateSocketA
- api | ws2_32: WSACleanup
- api | advapi32: GetLengthSid
- api | advapi32: RegOpenKeyA
- api | user32: SetForegroundWindow
- api | user32: GetWindowLongA
- api | user32: TrackPopupMenu
- api | user32: ShowCaret
- api | user32: InvalidateRect
- api | user32: GetClientRect
- api | user32: EnableWindow
- api | user32: UpdateWindow
- api | kernel32: GetStartupInfoA
- api | kernel32: GetModuleHandleA
- api | kernel32: GetModuleFileNameA
- api | kernel32: CreateFileA
- api | kernel32: FindClose
- api | kernel32: SetCurrentDirectoryW

- api | kernel32: CreateProcessW
- api | kernel32: VirtualFree
- api | kernel32: CreateFileMappingA
- api | kernel32: GetOverlappedResult
- api | kernel32: VirtualQuery
- api | kernel32: IsDBCSLeadByteEx
- api | kernel32: GetEnvironmentStrings
- api | kernel32: MulDiv
- api | ntdll: _ftol
- api | kernel32: VirtualAlloc
- api | kernel32: VirtualProtect
- api | kernel32: LoadLibraryA
- api | kernel32: lstrcpyA,(
- api | user32: CharUpperA
- api | kernel32: WaitForSingleObject
- api | kernel32: GetTempPathA
- api | kernel32: WaitForMultipleObjects
- api | kernel32: CreateProcessA
- api | kernel32: GetCurrentProcessId
- api | kernel32: GetCurrentProcess
- api | kernel32: CloseHandle
- api | kernel32: lstrcmpiA
- api | kernel32: lstrcatA
- api | kernel32: EnterCriticalSection
- api | kernel32: LeaveCriticalSection
- api | kernel32: InitializeCriticalSection
- api | kernel32: GetProcessHeap
- api | kernel32: HeapFree
- api | kernel32: HeapAlloc
- api | kernel32: lstrlenA
- api | kernel32: ResetEvent
- api | kernel32: SetEvent
- api | kernel32: OpenEventA
- api | kernel32: CreateEventA
- api | ntdll: ZwUnmapViewOfSection
- api | kernel32: IsWow64Process
- api | psapi: GetProcessImageFileNameA
- api | kernel32: GetExitCodeThread
- api | kernel32: RtlMoveMemory
- api | psapi: GetModuleBaseNameA
- api | kernel32: VirtualFreeEx

Involved hosts

- 88.135.242.207:http
- 192.168.133.1:5355
- 192.168.133.211:56889
- 192.168.133.211:57124

Dns queries

- 2.133.168.192.in-addr.arpa ---> no answers
- 207.242.135.88.in-addr.arpa ---> no answers
- dns.msftncsi.com ---> 131.107.255.255

Connections and readable network traffic

The analysis environment captures network traffic happened while sample execution. It classifies the traffic and tries to extract the most significant parts from the communications data.

In this section it's possible to find parts of the communication data that are easier to be read.

- **udp 192.168.133.1:5355 ---> 192.168.133.211:57124**

  in-addr[...]arpa[...]in-addr

- **udp 192.168.133.1:5355 ---> 192.168.133.211:56889**

  arpa

Threads behavioural information

In this section it's possible to find information about sample's threads, such as the actions performed by each sample's thread ordered chronologically.

• Thread T0 (in process P0, cebo.exe) description

• Thread's childs

• Thread T1 (in process P0, cebo.exe)

• Thread' events

• Thread Create (Thread ID: T1)

• Thread T1 (in process P0, cebo.exe) description

• Thread's childs

• Thread T2 (in process P1, cebo.exe)

• Thread' events

• RegCreateKey (HKCU\\Software\\Local AppWizard-Generated Applications Desired Access: Read/Write, Disposition: REG_CREATED_NEW_KEY)
• RegCreateKey (HKCU\\Software\\Local AppWizard-Generated Applications\\cebo Desired Access: Read/Write, Disposition: REG_CREATED_NEW_KEY)
• RegCreateKey (HKCU\\Software\\Local AppWizard-Generated Applications\\cebo\\Recent File List Desired Access: Read/Write, Disposition: REG_CREATED_NEW_KEY)
• RegCreateKey (HKCU\\Software\\Local AppWizard-Generated Applications Desired Access: Read/Write, Disposition: REG_OPENED_EXISTING_KEY)
• RegCreateKey (HKCU\\Software\\Local AppWizard-Generated Applications\\cebo Desired Access: Read/Write, Disposition: REG_OPENED_EXISTING_KEY)
• RegCreateKey (HKCU\\Software\\Local AppWizard-Generated Applications\\cebo\\Recent File List Desired Access: Read/Write, Disposition: REG_OPENED_EXISTING_KEY)
• RegCreateKey (HKCU\\Software\\Local AppWizard-Generated Applications Desired Access: Read/Write, Disposition: REG_OPENED_EXISTING_KEY)
• RegCreateKey (HKCU\\Software\\Local AppWizard-Generated Applications\\cebo Desired Access: Read/Write, Disposition: REG_OPENED_EXISTING_KEY)
• RegCreateKey (HKCU\\Software\\Local AppWizard-Generated Applications\\cebo\\Recent File List Desired Access: Read/Write, Disposition: REG_OPENED_EXISTING_KEY)
• RegCreateKey (HKCU\\Software\\Local AppWizard-Generated Applications Desired Access: Read/Write, Disposition: REG_OPENED_EXISTING_KEY)
• RegCreateKey (HKCU\\Software\\Local AppWizard-Generated Applications\\cebo Desired Access: Read/Write, Disposition: REG_OPENED_EXISTING_KEY)
• RegCreateKey (HKCU\\Software\\Local AppWizard-Generated Applications\\cebo\\Recent File List Desired Access: Read/Write, Disposition: REG_OPENED_EXISTING_KEY)
• RegCreateKey (HKCU\\Software\\Local AppWizard-Generated Applications Desired Access: Read/Write, Disposition: REG_OPENED_EXISTING_KEY)
• RegCreateKey (HKCU\\Software\\Local AppWizard-Generated Applications\\cebo Desired Access: Read/Write, Disposition: REG_OPENED_EXISTING_KEY)
• RegCreateKey (HKCU\\Software\\Local AppWizard-Generated Applications\\cebo\\Settings Desired Access: Read/Write, Disposition: REG_CREATED_NEW_KEY)
• Process Create (C:\\Users\\javi\\Downloads\\cebo.exe PID: P1, Command line: "C:\\Users\\javi\\Downloads\\cebo.exe")

• Thread T2 (in process P1, cebo.exe) description

• Thread's childs

• Thread T3 (in process P1, cebo.exe)

- **Thread' events**
  - Thread Create (Thread ID: T3)

- **Thread T3 (in process P1, cebo.exe) description**
  - **Thread's childs**
    - Thread T4 (in process P1, cebo.exe)
  - **Thread' events**
    - Thread Create (Thread ID: T4)

- **Thread T4 (in process P1, cebo.exe) description**
  - **Thread's childs**
    - Thread T5 (in process P1, cebo.exe)
  - **Thread' events**
    - Thread Create (Thread ID: T5)
    - Thread Create (Thread ID: TUNKALIAS)

- **Thread T5 (in process P1, cebo.exe) description**
  - **Thread's childs**
    - Thread T6 (in process P1, cebo.exe)
  - **Thread' events**
    - Thread Create (Thread ID: T6)

- **Thread T6 (in process P1, cebo.exe) description**
  - **Thread's childs**
    - Thread T7 (in process P1, cebo.exe)
  - **Thread' events**
    - Thread Create (Thread ID: T7)

- **Thread T7 (in process P1, cebo.exe) description**
  - **Thread's childs**
  - **Thread' events**

• Thread Create (Thread ID: TUNKALIAS)

Malware processes' network events

The analysis environment tries to capture and collect network actions performed by sample's threads.

Process P1 (cebo.exe)'s network events

- TCP Reconnect (javi-PC.localdomain:dynport-> 88.135.242.207:http (0))
- TCP Reconnect (javi-PC.localdomain:dynport-> 88.135.242.207:http (0))

Sample's unpacked / injected modules

In this section it's possible to find information about sample's modules, such as the rich signatures, strings, yara rules matching the module,...

The analysis environment performs an heuristic search trying to find sample's injected or unpacked modules in memory, dumping and analyzing them.

• Module 1 (probably unpacked / injected by the sample)

• Module 1 rich signatures

• 44616e5300bc00000000000000000000c30f5d0008000000831c0e00020000006f1f0a000b0000006f1f04000200000036260b000300000036260400030000
0000000100f7000000e81f0b0005000000c7060600

• Module 1 yara matches

• Microsoft_Visual_Cpp_v50v60_MFC ()
• IsPE32 ()
• IsWindowsGUI ()
• ImportTableIsBad (ImportTable Check)
• HasRichSignature (Rich Signature Check)
• without_attachments (Rule to detect the no presence of any attachment)
• without_images (Rule to detect the no presence of any image)
• without_urls (Rule to detect the no presence of any url)
• contentis_base64 (This rule finds for base64 strings)

• Module 1 strings

• Module 1 most interesting strings

• api | frequently used by malware: GetTimeZoneInformation
• api | frequently used by malware: GetTickCount
• api | advapi32: GetLengthSid
• api | advapi32: RegOpenKeyA
• api | user32: SetForegroundWindow
• api | user32: GetWindowLongA
• api | user32: TrackPopupMenu
• api | user32: ShowCaret
• api | user32: InvalidateRect
• api | user32: GetClientRect
• api | user32: EnableWindow
• api | user32: UpdateWindow
• api | kernel32: GetStartupInfoA
• api | kernel32: GetModuleHandleA
• api | kernel32: GetModuleFileNameA
• api | kernel32: CreateFileA
• api | kernel32: FindClose
• api | kernel32: SetCurrentDirectoryW
• api | kernel32: CreateProcessW
• api | kernel32: VirtualFree
• api | kernel32: CreateFileMappingA
• api | kernel32: GetOverlappedResult
• api | kernel32: VirtualQuery
• api | kernel32: IsDBCSLeadByteEx
• api | kernel32: GetEnvironmentStrings
• api | kernel32: MulDiv

- api | ntdll: _ftol

• Module 1 other strings

- ???? (C) 2007
- ImageViewer 1.0 ?
- ?? ImageViewer
- ?? ImageViewer(&A;)...
- ??25%
- ??50%
- ??25%
- ??50%
- wwp8
- OGp888888
- p884
- 33330p333333
- p884
- 3333
- 33330wp3
- p884
- 3333
- wwp3
- CMainFrame
- %K722N
- AF5@H-s22
- 622N
- @H-s22
- -s22
- 222z
- 222t
- @p,s22EP-s22>f
- 222tR
- 222F5
- 222tL
- 222@
- 222@
- 1F=@
- %T122
- %'222
- 5GIX5622
- 2@IP5622D
- FYP5622D
- T:6G
- F<T:3F7FIp5622@yp5622
- 2@IP5622
- FYP5622D
- T:6G
- 2@x5622F
- @H5622t
- CImageViewerView
- erDoc
- CImageVie
- Local AppWizard-Generated Applications
- ADVAPI32.dll
- GDI32.dll
- IntersectClipRect

- GetTextExtentExPointW
- SetMapMode
- GetDeviceCaps
- CreateCompatibleDC
- CreateCompatibleBitmap
- SelectObject
- USER32.dll
- KERNEL32.dll
- _controlfp
- _onexit
- __dllonexit
- _except_handler3
- __set_app_type
- __p__fmode
- __p__commode
- _adjust_fdiv
- __setusermatherr
- _initterm
- __getmainargs
- exit
- _exit
- __CxxFrameHandler
- MFC42.DLL
- %x3@
- >"u:F
- 5TW@
- %|2@
- %x2@
- %l2@
- %h2@
- %d2@
- %`2@
- %\\2@
- %X2@
- %T2@
- %P2@
- %L2@
- %H2@
- %D2@
- %@2@
- %<2@
- %82@
- %42@
- %02@
- %,2@
- %(2@
- %$2@
- % 2@
- %p3@
- %|1@
- %x1@
- %t1@
- %p1@
- %l1@
- %\\1@
- %X1@
- %T1@

- %P1@
- %L1@
- %H1@
- %D1@
- %@1@
- %<1@
- %81@
- %41@
- %01@
- %,1@
- %(1@
- %$1@
- % 1@
- %|0@
- %x0@
- %t0@
- % 3@
- %$3@
- %(3@
- %,3@
- %03@
- %43@
- %83@
- %<3@
- %@3@
- %D3@
- %H3@
- %L3@
- %P3@
- %T3@
- %X3@
- %\\3@
- %`3@
- %h3@
- %l3@
- %t3@
- uFPYt
- h4(@P
- D$0P
- D$0P
- T$8PR
- T$8PR
- L$8PQ
- -<0@
- TSUV
- h 6@
- @.data
- `.rdata
- .teyt
- 8RichW
- !This program cannot be run in DOS mode.
- ??(&H;)
- ????
- ??(&O;)
- ???(&S;)
- ???(&T;)
- ??(&V;)

- ??(&P;)\tCtrl+V
- ??(&C;)\tCtrl+C
- ??(&T;)\tCtrl+X
- ??(&U;)\tCtrl+Z
- ??(&E;)
- ??(&X;)
- ????
- ????(&R;)...
- ????(&V;)
- ??(&P;)...\tCtrl+P
- ???(&A;)...
- ??(&S;)\tCtrl+S
- ??(&O;)...\tCtrl+O
- ??(&N;)\tCtrl+N
- ??(&F;)
- @jjj
- @jjjj
- @jjjjjjj
- @jjj
- @jjjjjjj
- @jjjjjj
- @jjj
- @jjjjjjjjj
- jjjj
- wwwp
- wwwwwp
- wwwwwww
- wwwwwww
- wwwp
- wwwww
- wwwwwwpwppp0wwww
- wwwp
- p0wwww
- wwwwwwww
- DDGwp8
- p0wwww
- wwwp
- wwwppwwwp
- wwp0wwww
- wwwp
- ppww
- wwwp
- wwwp
- wwp0wwww
- wwwp
- wwwp
- wwww
- wwp0wwww
- wwwwp
- wwwwpppwww
- wwwp
- wwwt
- wwwwp
- pwww
- wwwttDwwp
- wwwwp
- wwtDtwGwp

- wwwwp
- wwwp
- wwGttwGwwwwt
- wwwp
- wwGttwGwwwwt
- wwwp
- pwwwwppwwww
- wwGtwDwwwwwtDDDDw
- wwwwp
- wwwwp
- wwtGwwwwwwwwwwwwwwwtDDDDw
- wwwwwwwwwwwwwwww

- h P@
- hR)@
- .rsrc

• Module 2 (probably unpacked / injected by the sample)

• Module 2 rich signatures

- 44616e5300bc0000000000000000000000c30f5d00080000000831c0e00020000006f1f0a000b0000006f1f04000200000036260b0003000000036260400030000000000100f7000000e81f0b0005000000c7060600

• Module 2 yara matches

- Microsoft_Visual_Cpp_v50v60_MFC ()
- IsPE32 ()
- IsWindowsGUI ()
- ImportTableIsBad (ImportTable Check)
- HasRichSignature (Rich Signature Check)
- without_attachments (Rule to detect the no presence of any attachment)
- without_images (Rule to detect the no presence of any image)
- without_urls (Rule to detect the no presence of any url)
- contentis_base64 (This rule finds for base64 strings)

• Module 2 strings

• Module 2 most interesting strings

- api | frequently used by malware: GetTimeZoneInformation
- api | frequently used by malware: GetTickCount
- api | advapi32: GetLengthSid
- api | advapi32: RegOpenKeyA
- api | user32: SetForegroundWindow
- api | user32: GetWindowLongA
- api | user32: TrackPopupMenu
- api | user32: ShowCaret
- api | user32: InvalidateRect
- api | user32: GetClientRect
- api | user32: EnableWindow
- api | user32: UpdateWindow
- api | kernel32: GetStartupInfoA
- api | kernel32: GetModuleHandleA
- api | kernel32: GetModuleFileNameA
- api | kernel32: CreateFileA
- api | kernel32: FindClose
- api | kernel32: SetCurrentDirectoryW
- api | kernel32: CreateProcessW
- api | kernel32: VirtualFree
- api | kernel32: CreateFileMappingA
- api | kernel32: GetOverlappedResult
- api | kernel32: VirtualQuery
- api | kernel32: IsDBCSLeadByteEx
- api | kernel32: GetEnvironmentStrings
- api | kernel32: MulDiv
- api | ntdll: _ftol

• Module 2 other strings

• ???? (C) 2007
• ImageViewer 1.0 ?
• ?? ImageViewer
• ?? ImageViewer(&A;)...
• ??25%
• ??50%
• ??25%
• ??50%
• wwp8
• OGp888888
• p884
• 33330p333333
• p884
• 3333
• 33330wp3
• p884
• 3333
• wwp3
• CMainFrame
• %K722N
• AF5@H-s22
• 622N
• @H-s22
• -s22
• 222z
• 222t
• @p,s22EP-s22>f
• 222tR
• 222F5
• 222tL
• 222@
• 222@
• 1F=@
• %T122
• %'222
• 5GIX5622
• 2@IP5622D
• FYP5622D
• T:6G
• F<T:3F7FIp5622@yp5622
• 2@IP5622
• FYP5622D
• T:6G
• 2@x5622F
• @H5622t
• CImageViewerView
• erDoc
• CImageVie
• Local AppWizard-Generated Applications
• ADVAPI32.dll
• GDI32.dll
• IntersectClipRect
• GetTextExtentExPointW
• SetMapMode

- GetDeviceCaps
- CreateCompatibleDC
- CreateCompatibleBitmap
- SelectObject
- USER32.dll
- KERNEL32.dll
- _controlfp
- _onexit
- __dllonexit
- _except_handler3
- __set_app_type
- __p__fmode
- __p__commode
- _adjust_fdiv
- __setusermatherr
- _initterm
- __getmainargs
- exit
- _exit
- __CxxFrameHandler
- MFC42.DLL
- %x3@
- >"u:F
- 5TW@
- %|2@
- %x2@
- %l2@
- %h2@
- %d2@
- %`2@
- %\\2@
- %X2@
- %T2@
- %P2@
- %L2@
- %H2@
- %D2@
- %@2@
- %<2@
- %82@
- %42@
- %02@
- %,2@
- %(2@
- %$2@
- % 2@
- %p3@
- %|1@
- %x1@
- %t1@
- %p1@
- %l1@
- %\\1@
- %X1@
- %T1@
- %P1@
- %L1@

- %H1@
- %D1@
- %@1@
- %<1@
- %81@
- %41@
- %01@
- %,1@
- %(1@
- %$1@
- % 1@
- %|0@
- %x0@
- %t0@
- % 3@
- %$3@
- %(3@
- %,3@
- %03@
- %43@
- %83@
- %<3@
- %@3@
- %D3@
- %H3@
- %L3@
- %P3@
- %T3@
- %X3@
- %\\3@
- %`3@
- %h3@
- %l3@
- %t3@
- uFPYt
- h4(@P
- D$0P
- D$0P
- T$8PR
- T$8PR
- L$8PQ
- -<0@
- TSUV
- h 6@
- @.data
- `.rdata
- .teyt
- 8RichW
- !This program cannot be run in DOS mode.
- ??(&H;)
- ????
- ??(&O;)
- ???(&S;)
- ???(&T;)
- ??(&V;)
- ??(&P;)\tCtrl+V
- ??(&C;)\tCtrl+C

- ??(&T;)\tCtrl+X
- ??(&U;)\tCtrl+Z
- ??(&E;)
- ??(&X;)
- ????
- ????(&R;)...
- ????(&V;)
- ??(&P;)...\tCtrl+P
- ???(&A;)...
- ??(&S;)\tCtrl+S
- ??(&O;)...\tCtrl+O
- ??(&N;)\tCtrl+N
- ??(&F;)
- @jjj
- @jjjj
- @jjjjjjj
- @jjj
- @jjjjjjj
- @jjjjjj
- @jjj
- @jjjjjjjjj
- jjjj
- wwwp
- wwwwwp
- wwwwwww
- wwwwwww
- wwwp
- wwwww
- wwwwwwpwppp0wwww
- wwwp
- p0wwww
- wwwwwwww
- DDGwp8
- p0wwww
- wwwp
- wwwppwwwp
- wwp0wwww
- wwwp
- ppww
- wwwp
- wwwp
- wwp0wwww
- wwwp
- wwwp
- wwww
- wwp0wwww
- wwwwp
- wwwwpppwww
- wwwp
- wwwt
- wwwwp
- pwww
- wwwttDwwp
- wwwwp
- wwtDtwGwp
- wwwwp
- wwwp

- wwGttwGwwwwt
- wwwp
- wwGttwGwwwwt
- wwwp
- pwwwwppwwww
- wwGtwDwwwwwtDDDDw
- wwwwp
- wwwwp
- wwtGwwwwwwwwwwwwwwwtDDDDw
- wwwwwwwwwwwwwwww

- hh)@
- L$(d
- h P@
- hR)@
- .rsrc

• Module 3 (probably unpacked / injected by the sample)

• Module 3 rich signatures

- 44616e5300bc0000000000000000000000c30f5d00080000000831c0e00020000006f1f0a000b0000006f1f04000200000036260b0003000000036260400030000000000100f7000000e81f0b0005000000c7060600

• Module 3 yara matches

- Microsoft_Visual_Cpp_v50v60_MFC ()
- IsPE32 ()
- IsWindowsGUI ()
- ImportTableIsBad (ImportTable Check)
- HasRichSignature (Rich Signature Check)
- without_attachments (Rule to detect the no presence of any attachment)
- without_images (Rule to detect the no presence of any image)
- without_urls (Rule to detect the no presence of any url)
- contentis_base64 (This rule finds for base64 strings)

• Module 3 strings

• Module 3 most interesting strings

- api | frequently used by malware: GetTimeZoneInformation
- api | frequently used by malware: GetTickCount
- api | advapi32: GetLengthSid
- api | advapi32: RegOpenKeyA
- api | user32: SetForegroundWindow
- api | user32: GetWindowLongA
- api | user32: TrackPopupMenu
- api | user32: ShowCaret
- api | user32: InvalidateRect
- api | user32: GetClientRect
- api | user32: EnableWindow
- api | user32: UpdateWindow
- api | kernel32: GetStartupInfoA
- api | kernel32: GetModuleHandleA
- api | kernel32: GetModuleFileNameA
- api | kernel32: CreateFileA
- api | kernel32: FindClose
- api | kernel32: SetCurrentDirectoryW
- api | kernel32: CreateProcessW
- api | kernel32: VirtualFree
- api | kernel32: CreateFileMappingA
- api | kernel32: GetOverlappedResult
- api | kernel32: VirtualQuery
- api | kernel32: IsDBCSLeadByteEx
- api | kernel32: GetEnvironmentStrings
- api | kernel32: MulDiv

• api | ntdll: _ftol

• Module 3 other strings

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• OGp888888
• p884
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• 33330wp3
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• 3333
• wwp3
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• 622N
• @H-s22
• -s22
• 222z
• 222t
• @p,s22EP-s22>f
• 222tR
• 222F5
• 222tL
• 222@
• 222@
• 1F=@
• %T122
• %'222
• 5GIX5622
• 2@IP5622D
• FYP5622D
• T:6G
• F<T:3F7FIp5622@yp5622
• 2@IP5622
• FYP5622D
• T:6G
• 2@x5622F
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• CImageViewerView
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• CImageVie
• Local AppWizard-Generated Applications
• ADVAPI32.dll
• GDI32.dll
• IntersectClipRect

- GetTextExtentExPointW
- SetMapMode
- GetDeviceCaps
- CreateCompatibleDC
- CreateCompatibleBitmap
- SelectObject
- USER32.dll
- KERNEL32.dll
- _controlfp
- _onexit
- __dllonexit
- _except_handler3
- __set_app_type
- __p__fmode
- __p__commode
- _adjust_fdiv
- __setusermatherr
- _initterm
- __getmainargs
- exit
- _exit
- __CxxFrameHandler
- MFC42.DLL
- %x3@
- >"u:F
- 5TW@
- %|2@
- %x2@
- %l2@
- %h2@
- %d2@
- %`2@
- %\\2@
- %X2@
- %T2@
- %P2@
- %L2@
- %H2@
- %D2@
- %@2@
- %<2@
- %82@
- %42@
- %02@
- %,2@
- %(2@
- %$2@
- % 2@
- %p3@
- %|1@
- %x1@
- %t1@
- %p1@
- %l1@
- %\\1@
- %X1@
- %T1@

- %P1@
- %L1@
- %H1@
- %D1@
- %@1@
- %<1@
- %81@
- %41@
- %01@
- %,1@
- %(1@
- %$1@
- % 1@
- %|0@
- %x0@
- %t0@
- % 3@
- %$3@
- %(3@
- %,3@
- %03@
- %43@
- %83@
- %<3@
- %@3@
- %D3@
- %H3@
- %L3@
- %P3@
- %T3@
- %X3@
- %\\3@
- %`3@
- %h3@
- %l3@
- %t3@
- uFPYt
- h4(@P
- D$0P
- D$0P
- T$8PR
- T$8PR
- L$8PQ
- -<0@
- TSUV
- h 6@
- @.data
- `.rdata
- .teyt
- 8RichW
- !This program cannot be run in DOS mode.
- ??(&H;)
- ????
- ??(&O;)
- ???(&S;)
- ???(&T;)
- ??(&V;)

- ??(&P;)\tCtrl+V
- ??(&C;)\tCtrl+C
- ??(&T;)\tCtrl+X
- ??(&U;)\tCtrl+Z
- ??(&E;)
- ??(&X;)
- ????
- ????(&R;)...
- ????(&V;)
- ??(&P;)...\tCtrl+P
- ???(&A;)...
- ??(&S;)\tCtrl+S
- ??(&O;)...\tCtrl+O
- ??(&N;)\tCtrl+N
- ??(&F;)
- @jjj
- @jjjj
- @jjjjjjj
- @jjj
- @jjjjjjj
- @jjjjjj
- @jjj
- @jjjjjjjjj
- jjjj
- wwwp
- wwwwwp
- wwwwwww
- wwwwwww
- wwwp
- wwwww
- wwwwwwpwppp0wwww
- wwwp
- p0wwww
- wwwwwwww
- DDGwp8
- p0wwww
- wwwp
- wwwppwwwp
- wwp0wwww
- wwwp
- ppww
- wwwp
- wwwp
- wwp0wwww
- wwwp
- wwwp
- wwww
- wwp0wwww
- wwwwp
- wwwwpppwww
- wwwp
- wwwt
- wwwwp
- pwww
- wwwttDwwp
- wwwwp
- wwtDtwGwp

- wwwwp
- wwwp
- wwGttwGwwwwt
- wwwp
- wwGttwGwwwwt
- wwwp
- pwwwwppwwww
- wwGtwDwwwwwtDDDDw
- wwwwp
- wwwwp
- wwtGwwwwwwwwwwwwwwwtDDDDw
- wwwwwwwwwwwwwwww

- h P@
- hR)@
- .rsrc

• Module 4 (probably unpacked / injected by the sample)

• Module 4 rich signatures

- 44616e530000000000000000000000000027c67b0005000000000001001d0000000097884000200000009788a000900000009789100

• Module 4 yara matches

- suspicious_packer_section (The packer/protector section names/keywords)
- UPX ()
- Str_Win32_Winsock2_Library (Match Winsock 2 API library declaration)
- Str_Win32_Wininet_Library (Match Windows Inet API library declaration)
- Str_Win32_Internet_API (Match Windows Inet API call)
- PackerUPX_CompresorGratuito_wwwupxsourceforgenet ()
- UPX_wwwupxsourceforgenet_additional ()
- yodas_Protector_v1033_dllocx_Ashkbiz_Danehkar_h ()
- Netopsystems_FEAD_Optimizer_1 ()
- UPX_290_LZMA ()
- UPX_290_LZMA_Markus_Oberhumer_Laszlo_Molnar_John_Reiser ()
- UPX_290_LZMA_additional ()
- UPX_wwwupxsourceforgenet ()
- IsPE32 ()
- IsWindowsGUI ()
- HasOverlay (Overlay Check)
- HasRichSignature (Rich Signature Check)
- UPXv20MarkusLaszloReiser ()
- UPXV200V290MarkusOberhumerLaszloMolnarJohnReiser ()
- UPX20030XMarkusOberhumerLaszloMolnarJohnReiser ()
- UPX290LZMAMarkusOberhumerLaszloMolnarJohnReiser ()
- upx_3 (UPX 3.X)
- without_attachments (Rule to detect the no presence of any attachment)
- without_images (Rule to detect the no presence of any image)
- without_urls (Rule to detect the no presence of any url)
- contentis_base64 (This rule finds for base64 strings)

• Module 4 strings

• Module 4 most interesting strings

- url | valid domains | tlds: .ms-*nl
- url | valid domains | tlds: 7@.Tw
- api | frequently used by malware: GetNetworkParams
- api | frequently used by malware: ExitProcess
- api | frequently used by malware: GetProcAddress
- api | frequently used by malware: Send
- api | frequently used by malware: ExitProcess
- api | frequently used by malware: GetProcAddress
- api | wininet: InternetOpenA
- api | kernel32: VirtualFree
- api | kernel32: VirtualAlloc
- api | kernel32: VirtualProtect

- api | kernel32: LoadLibraryA
- api | kernel32: lstrcpyA,(
- api | user32: CharUpperA
- api | kernel32: VirtualFree
- api | kernel32: VirtualAlloc
- api | kernel32: VirtualProtect
- api | kernel32: LoadLibraryA

• Module 4 other strings

- &Fil;
- ShellExecuteA
- time
- WS2_32.dll
- WININET.dll
- SHELL32.dll
- OLEAUT32.dll
- MFC42.DLL
- iphlpapi.dll
- KERNEL32.DLL
- Pa5,`@
- SAIo
- :D0aAva-
- dERff
- _RequefHCv
- 7lenv
- zoi
- +fdiv
- J\\DO
- Raise
- _>ModuHandl!O
- fe W
- 7Infoa
- 9 \\D
- pP;h4riCM
- Libra
- Addr(
- ermind<
- All-
- e]LaE[{
- oGtualF
- ExiBt
- ctor
- `\tatA!t
- ckCo
- T6FlAg;fa
- {1tc
- Ae+H
- M1\t9e\t{
- osssygd1
- 6{o0+e
- l5Qb1k[u
- a%skE
- i;q{
- 27v`
- 3w#|I 9f

- 6=10;
- v-=3
- dI:!e
- y5tLe/
- a=g;I1o
- g}wa0
- S 2D
- `+du
- uoB? | CO
- 5-06
- IPTION\\S[tm-
- oft\\
- OFTWARE\\M
- \\qaz
- URLD
- wordZ
- jgif,
-)[A6
- {O:"y1)
- -Ag=:
- TP/E
- -En+d
- Accep
- 82p6
- 7_0K
- 'iPadqCPU
- SS15Q
- [9Fa.
- GT-I9
- `l5E*){H ?2&7.367
- Q9B0\\
- 6C1dL
- }0727)K8
- i-[4}
- afe
- 3n253KHTML
- [wKi
- :_8)V
- OS X
- KMac
- uJ{)
- WOWuTv,
- v26'J3
- MSIE 9
- efox
- rv:16#)aecko]o
- ux x86
- ; IX13
-)3.6
- 2.1Z+A
- 41/s,)
- /yse
- Slurp
- Yahoo!
- ;%7{
- /5u (
- 89,M

- /web
- ainMk
- */*; q01C+
- defl
- gzip?1.0*w
- 5f-!
- en-UK;q=0.8,[
- about:blank
- 4hP:D
- pd4
- eWno
- \\7\t.
- es\\InFrMt ExIo
- F.meJ\\Prog
- tions+CMy1D
- l Ap*zard-Gener
- Loca
- OpenTd
- Qu`yVal
- RegCl)eKey
- #InputS
- ?Thread
- KiCTimer[
- essageA
- PostM
- Enabld
- {tfA
- pdateWindowwspr
- \\`qr3
- +USER32.dll
- ,O&O;
- B^DO>
- i@:4.(
- 4B<60N{M
- iztnP
- ~xrli
- X3u;
- Q$cu
- <4Rfd
- -0xl
- B`.3
- >"u:F
- T\t`e
- 0b4b8b<b@b
- 048D
- |xt2222plhd2222`\\XT
- 2222
- 2222
- 2222
- 2222
- 2226
- <222
- qhYI
- qlDTu
- Zv\t+u
- !g<h8,8
- pYoa^

- kt13d
- 3d#%&
- t*HuX
- %a3x
- +9E`
- ^a/~
- 7B"9
- v(:9
- 6Y(1tr
- x8u2!y
- l.$E
- (n5r!
- +0p"\\
- 5Qmb
- 1IAW
- rra8k
- P UQL
- W @E
- 4L@S
- re<
- W3lx
- XXvI
- yF!EQ
- j UW
- R90`4
- N:c7
- ?zu
- c;t0
- a0&9
- 5>3z
- C@1E9
- h XU
- 7GD;
- VU7D
- TS0
- 8 $fR
- \\8%5
- ;3[B
- aWYrH
- `xLO
- 5HWY
- k2Sj
- }1g|
- 7CF(;
- _(l8u
- BBIu
- q?pa
- h?ji
- uI01*^Q
- wB<a
- yx2?
- w A@
- (SV3}
- '{@2Y
- UPX!
- 3.91
- UPX1

- UPX0
-]Rich
- !This program cannot be run in DOS mode.
- USER32.dll
- KERNEL32.DLL
- .s2'B
- OeWS
- C.xU
- n,hA5
- / AEGCuraW
- Addsss;oadLibraryA
- E`)[
- l;X2
- InitializeCr\tc
- kCount
- 0Tic
- HeapK
- *[TU
- :F8
- CH!3_o2
- SiBFObj
- dp>ewOfSec
- ZwUn6
- 6v1FD.S=
- 200 OK
- gthIo
- m-urAnczdT
- #?'1Ha?
- /a4s
- ET 3 HTTP/1.0
- Ws2_
- 32.Dl
- k>nel
- IsWow64*
- ImageFilu/<
- Opens
- veSCu7wk
- NamMl
- Enum
- cessMemyy
- EGetModuleHand
- Protec
- F"eEx
- Virtu0A
- rer.m
- bad allocation
- 9y&9
- 4kqw}
- 5;AM
- 1@22
- \\A5l
- fwixM
- U]Qd
- tH\\4f0
- IZC0J
- '8b.
- P[3e

- 8-h,[
- *a3E
- !]dCk10A
- K}9[6
-],8P
- <)6
- 8B@'
- VjD3
- o3pw
- 5\\f!
- ~1BOK
- `YWOj
- maV%
- 2jQu
- 22S2
- $$(4,
- X4iN
- t4Hu
- L7|1
- 1uKI
- uo&v;|
- UVWhXC3
- KXq90
- UPX!
- 3.91
- UPX2
- UPX1
- UPX0
- C.xU
- Rich8
- !This program cannot be run in DOS mode.
- w\tCt
- MSVCRT.dll
- XPTPSW
- # :.rd}!
- mCxx
- yrgs9cmnn
- _fkP
- LPm[
- m{/l
- kPW
- ked{ch
- b(J$
- T-X#
- TkLm
- @*M_&e;#!
- sz!-&{
- SeKp\\%wq
-),.{_E
- -]p^
- $V #
- cP)k
- rkg
- hwc=
- a]R+CG
- c/mYJ&
- bS2kMc

- C{%
- FF&X;"o
- R%c)
- &{O,N&S;>
- E [+D/DESCR
- \\CFrC
- Z{KexY
- x-xbitp
- ssrt
-]\t){
- fmXi\tG
- -Typ!
- "v\t/
- /JDQR
- CLRs+
- kek&V.;
- osh?~W
- s NT 6
- * G#g
- \\tp:/uelp.y
- ml$aGx
- ascr.t
- [k_/js
- sdch
- compPs
- 4zh-CN,
- CChild
- }ktop;
- GD&K;
- D' pd
- [W V@
- .htm0H
- GET ^&&%$
- d^XRLFi
- |vpj
- @.~N
- if`ZTN
- \\rr\\``8
- %'XX\\
- \\rhhTT
- %HLL
- \\PDDHr
- !QRhl
- mRhrlN
- xBJJ
- bb####bbcc####cccc####cccc####cccc####ccccQ###ccccT/
- |pbtbxb|b|>
- TbXb\\bhblb
- |>DbHbLbPb>
- b b$b(b,b
- |>GF
- hlpt
- X\\`d
- HLPT
- $(,
- #@Ll
- m@P`+

- PSh^
- ~>(`
- :LKz
- }L(Sh cT
- [_;f
- ~hLpg
- n[Fd
- fd*W
- `*Dl
- @Qb
- sm="x
- dj".
- p }Lkb
- wZ<^
- `nSN
- @KG(
- K\tZ>
- (#r=*2%,
- P>ESXnr'W
- @WSV!
- c;@n
-)j
- S/D+m
- <?pm
- t\th,^p
- wm!{r
- t=<n_
- fg*S+
- rrrr
- vj[]Y
- FK}
- v~Wd
- WM]NnD
- \\(c$$&l;
- <r%Wr$(,
- -Xf$
- -{z~
- (LXV
- \\HrY
- Y@P(
- l&Yv;
- q)P.
- P?Fh
- .rsrc
- XPTPSW
- }.rd
- *HFT,
- P$oLB@
- m@l(
- c{7>cv
- &CStwtO;
- /x-w
- -Ty@
- POST #+
- d%k[
- strcmpiXRtl
- 4MGMSY_e4M

- dddd,<P\\sdddl
- V;#
- \\Z*X
- bQuqV+
- D@N`
- }fHV
- AlcSBp\tv
- m3x",?
- P'k(
- [Q<
- '#bx
- _^yT
- SSBS'
- ,PSl
- slQH
- ^G@V
- SQR
- Rh~f
- (pw`K
- hDtH
- tMhD
- H>Qx
- h@;P/
- tj:QP<Z
- ,#VaGC
- h(xVM
- .\thd
- s(hmv
- FH"Q
- +'@9\ts#}
- ;JsD

• Module 5 (probably unpacked / injected by the sample)

• Module 5 rich signatures

- 44616e530000000000000000000000000027c67b0005000000000001001d0000000097884000200000009788a000900000009789100

• Module 5 yara matches

- suspicious_packer_section (The packer/protector section names/keywords)
- UPX ()
- Str_Win32_Winsock2_Library (Match Winsock 2 API library declaration)
- IsPE32 ()
- IsWindowsGUI ()
- HasOverlay (Overlay Check)
- HasRichSignature (Rich Signature Check)
- UPXv20MarkusLaszloReiser ()
- UPXV200V290MarkusOberhumerLaszloMolnarJohnReiser ()
- UPX20030XMarkusOberhumerLaszloMolnarJohnReiser ()
- inject_thread (Code injection with CreateRemoteThread in a remote process)
- network_tcp_listen (Listen for incoming communication)
- network_tcp_socket (Communications over RAW socket)
- network_dns (Communications use DNS)
- win_files_operation (Affect private profile)
- without_attachments (Rule to detect the no presence of any attachment)

- without_images (Rule to detect the no presence of any image)
- without_urls (Rule to detect the no presence of any url)
- contentis_base64 (This rule finds for base64 strings)

• Module 5 strings

• Module 5 most interesting strings

- url | valid domains | tlds: 3.vA#-v
- url | valid domains | tlds: 3.v\\9.vA
- api | frequently used by malware: ExitProcess
- api | frequently used by malware: GetProcAddress
- api | frequently used by malware: CreateThread
- api | frequently used by malware: WriteFile
- api | frequently used by malware: ExitProcess
- api | frequently used by malware: OpenProcess
- api | frequently used by malware: Sleep
- api | frequently used by malware: GetProcAddress
- api | frequently used by malware: GetTickCount
- api | frequently used by malware: recv
- api | frequently used by malware: send
- api | frequently used by malware: WSAStartup
- api | frequently used by malware: connect
- api | frequently used by malware: htons
- api | frequently used by malware: gethostbyname
- api | frequently used by malware: closesocket
- api | frequently used by malware: socket
- api | frequently used by malware: Sleep
- api | frequently used by malware: send
- api | frequently used by malware: recv
- api | frequently used by malware: WSAStartup
- api | frequently used by malware: socket
- api | frequently used by malware: gethostbyname
- api | frequently used by malware: htons
- api | frequently used by malware: connect
- api | frequently used by malware: closesocket
- api | frequently used by malware: OpenProcess
- api | frequently used by malware: CreateRemoteThread
- api | frequently used by malware: EnumProcesses
- api | frequently used by malware: WriteProcessMemory
- api | frequently used by malware: ReadProcessMemory
- api | frequently used by malware: VirtualAllocEx
- api | ws2_32: __WSAFDIsSet
- api | ws2_32: select
- api | ws2_32: WSASocketA
- api | ws2_32: WSAGetLastError
- api | ws2_32: ioctlsocket
- api | ws2_32: WSADuplicateSocketA
- api | ws2_32: __WSAFDIsSet
- api | ws2_32: select
- api | ws2_32: ioctlsocket
- api | ws2_32: WSAGetLastError
- api | ws2_32: WSACleanup
- api | user32: CharUpperA
- api | kernel32: VirtualFree
- api | kernel32: VirtualAlloc

- api | kernel32: VirtualProtect
- api | kernel32: LoadLibraryA
- api | user32: CharUpperA
- api | kernel32: WaitForSingleObject
- api | kernel32: GetTempPathA
- api | kernel32: WaitForMultipleObjects
- api | kernel32: CreateProcessA
- api | kernel32: CreateFileA
- api | kernel32: GetCurrentProcessId
- api | kernel32: GetCurrentProcess
- api | kernel32: CloseHandle
- api | kernel32: lstrcmpiA
- api | kernel32: lstrcatA
- api | kernel32: GetModuleHandleA
- api | kernel32: LoadLibraryA
- api | kernel32: EnterCriticalSection
- api | kernel32: LeaveCriticalSection
- api | kernel32: InitializeCriticalSection
- api | kernel32: GetProcessHeap
- api | kernel32: HeapFree
- api | kernel32: HeapAlloc
- api | kernel32: lstrlenA
- api | kernel32: WaitForSingleObject
- api | kernel32: ResetEvent
- api | kernel32: SetEvent
- api | kernel32: OpenEventA
- api | kernel32: CreateEventA
- api | kernel32: lstrlenA
- api | ntdll: ZwUnmapViewOfSection
- api | kernel32: IsWow64Process
- api | psapi: GetProcessImageFileNameA
- api | kernel32: CloseHandle
- api | kernel32: GetExitCodeThread
- api | kernel32: RtlMoveMemory
- api | kernel32: lstrcmpiA
- api | psapi: GetModuleBaseNameA
- api | kernel32: GetModuleHandleA
- api | kernel32: VirtualProtect
- api | kernel32: VirtualFree
- api | kernel32: VirtualAlloc
- api | kernel32: VirtualFreeEx

• Module 5 other strings

- USER32.dll
- KERNEL32.DLL
- \\9.v
- .s2'B
- OeWS
- C.xU
- n,hA5
- / AEGCuraW
- Addsss;oadLibraryA
- E`)[
- I;X2
- InitializeCr\tc

- kCount
- 0Tic
- HeapK
- *[TU
- :F8
- CH!3_o2
- SiBFObj
- dp>ewOfSec
- ZwUn6
- 6v1FD.S=
- 200 OK
- gthIo
- m-urAnczdT
- #?'1Ha?
- /a4s
- ET 3 HTTP/1.0
- Ws2_
- 32.Dl
- k>nel
- IsWow64*
- ImageFilu/<
- Opens
- veSCu7wk
- NamMl
- Enum
- cessMemyy
- EGetModuleHand
- Protec
- F"eEx
- Virtu0A
- rer.m
- bad allocation
- 9y&9
- 4kqw}
- 5;AM
- 1@22
- \\A5l
- fwixM
- U]Qd
- tH\\4f0
- IZC0J
- '8b.
- P[3e
- 8-h,[
- *a3E
- !|dCk10A
- K}9[6
-],8P
- <)6
- 8B@'
- VjD3
- o3pw
- 5\\f!
- ~1BOK
- `YWOj
- maV%
- 2jQu

- 22S2
- $$(4,
- X4iN
- t4Hu
- L7|1
- .reloc
- @.data
- `.rdata
- .text
- C.xU
- Running
- WinSock 2.0
- SI*f
- :F8
- Ws2_32
- Kernel32
- wsock32.dll
- wsock32.dll
- Content-Length:
- 200 OK
- Content-Length:
- Content-Type: application/x-www-form-urlencoded
- Host:
- HTTP/1.1
- POST
- Host:
- HTTP/1.1
- GET
- Host:
- HTTP/1.0
- GET
- bad allocation
- /arisx06.exe
- .exe
- temp
- Host:
- HTTP/1.0
- GET
- Psapi
- Ws2_32.dll
- wsock32.dll
- user32.dll
- kernel32.dll
- explorer.exe
- explorer.exe
- bad allocation
- -vej/v]7.v
- h0,@
- 50@@
- 8PQj
- D$8Pj
- Wh8B@
- 50@@
- TVjD3
- h0B@
- 50@@
- Y9|$

- 50@@
- t2S2
- h<E@
- h4E@
- h$E@
- h0D@
- SUVW
- t4Hu
- t7|1
- h4C@
- 1uKF
- 5 @@
- <)6
- 8B@'
- VjD3
- o3pw
- 5\\f!
- ~1BOK
- `YWOj
- maV%
- 2jQu
- 22S2
- $$(4,
- X4iN
- t4Hu
- L7|1
- 1uKI
- uo&v;|
- UVWhXC3
- KXq90
- UPX!
- 3.91
- `UPX2
- `UPX1
- UPX0
- C.xU
- Rich8
- !This program cannot be run in DOS mode.
- .vO!.v
- XPTPSW
- }.rd
- *HFT,
- P$oLB@
- m@l(
- c{7>cv
- &CStwtO;
- /x-w
- -Ty@
- POST #+
- d%k[
- strcmpiXRtl
- 4MGMSY_e4M
- dddd,<P\\sdddl
- V;#
- \\Z*X
- bQuqV+
- D@N`

- }fHV
- AlcSBp\tv
- m3x",?
- P'k(
- [Q<
- '#bx
- _^yT
- SSBS'
- ,PSl
- slQH
- ^G@V
- SQR
- Rh~f
- (pw`K
- hDtH
- tMhD
- H>Qx
- h@;P/
- -v!>-v
- ntdll.dll
- ntdll.dll
- HTTP/1.
- -vO!.v
- -vl"-v|
- Ht#Ht
- t%Vj
- D$ P
- D$ P
- hHB@
- D$ P
- SVWhp
- Y_^[
- h@B@
- PVVVVVV
- h(B@
- QQSVWh
- Ph~f
- XjPf
- htB@
- hlB@
- tNSj
- Rh~f
- F<_^][
- htD@
- hhD@
- hXD@
- hLD@
- tMhDD@
- = @@
- h<D@
- =(@@
- hHC@
- h@C@
- h<C@
- tjWVQP
- _^[]
- h(C@

- h|C@
- hpC@
- u\thdC@
- -$@@
- UVWhXC@
- '#bx
- _^yT
- SSBS'
- ,PSl
- slQH
- ^G@V
- SQR
- Rh~f
- (pw`K
- hDtH
- tMhD
- H>Qx
- h@;P/
- tj:QP<Z
- ,#VaGC
- h(xVM
- .\thd
- s(hmv
- FH"Q
- +'@9\ts#}
- ;JsD

• Module 6 (probably unpacked / injected by the sample)

• Module 6 rich signatures

- 44616e530000000000000000000000000027c67b0005000000000001001d0000000097884000200000009788a000900000009789100

• Module 6 yara matches

- suspicious_packer_section (The packer/protector section names/keywords)
- UPX ()
- Str_Win32_Winsock2_Library (Match Winsock 2 API library declaration)
- IsPE32 ()
- IsWindowsGUI ()
- HasOverlay (Overlay Check)
- HasRichSignature (Rich Signature Check)
- UPXv20MarkusLaszloReiser ()
- UPXV200V290MarkusOberhumerLaszloMolnarJohnReiser ()
- UPX20030XMarkusOberhumerLaszloMolnarJohnReiser ()
- inject_thread (Code injection with CreateRemoteThread in a remote process)
- network_tcp_listen (Listen for incoming communication)
- network_tcp_socket (Communications over RAW socket)
- network_dns (Communications use DNS)
- win_files_operation (Affect private profile)
- without_attachments (Rule to detect the no presence of any attachment)
- without_images (Rule to detect the no presence of any image)
- without_urls (Rule to detect the no presence of any url)
- contentis_base64 (This rule finds for base64 strings)

- **Module 6 strings**

  - **Module 6 most interesting strings**

    - url | valid domains | tlds: 3.vA#-v
    - url | valid domains | tlds: 3.v\\9.vA
    - api | frequently used by malware: ExitProcess
    - api | frequently used by malware: GetProcAddress
    - api | frequently used by malware: CreateThread
    - api | frequently used by malware: WriteFile
    - api | frequently used by malware: ExitProcess
    - api | frequently used by malware: OpenProcess
    - api | frequently used by malware: Sleep
    - api | frequently used by malware: GetProcAddress
    - api | frequently used by malware: GetTickCount
    - api | frequently used by malware: recv
    - api | frequently used by malware: send
    - api | frequently used by malware: WSAStartup
    - api | frequently used by malware: connect
    - api | frequently used by malware: htons
    - api | frequently used by malware: gethostbyname
    - api | frequently used by malware: closesocket
    - api | frequently used by malware: socket
    - api | frequently used by malware: Sleep
    - api | frequently used by malware: send
    - api | frequently used by malware: recv
    - api | frequently used by malware: WSAStartup
    - api | frequently used by malware: socket
    - api | frequently used by malware: gethostbyname
    - api | frequently used by malware: htons
    - api | frequently used by malware: connect
    - api | frequently used by malware: closesocket
    - api | frequently used by malware: OpenProcess
    - api | frequently used by malware: CreateRemoteThread
    - api | frequently used by malware: EnumProcesses
    - api | frequently used by malware: WriteProcessMemory
    - api | frequently used by malware: ReadProcessMemory
    - api | frequently used by malware: VirtualAllocEx
    - api | ws2_32: __WSAFDIsSet
    - api | ws2_32: select
    - api | ws2_32: WSASocketA
    - api | ws2_32: WSAGetLastError
    - api | ws2_32: ioctlsocket
    - api | ws2_32: WSADuplicateSocketA
    - api | ws2_32: __WSAFDIsSet
    - api | ws2_32: select
    - api | ws2_32: ioctlsocket
    - api | ws2_32: WSAGetLastError
    - api | ws2_32: WSACleanup
    - api | user32: CharUpperA
    - api | kernel32: VirtualFree
    - api | kernel32: VirtualAlloc
    - api | kernel32: VirtualProtect
    - api | kernel32: LoadLibraryA
    - api | user32: CharUpperA
    - api | kernel32: WaitForSingleObject

- api | kernel32: GetTempPathA
- api | kernel32: WaitForMultipleObjects
- api | kernel32: CreateProcessA
- api | kernel32: CreateFileA
- api | kernel32: GetCurrentProcessId
- api | kernel32: GetCurrentProcess
- api | kernel32: CloseHandle
- api | kernel32: lstrcmpiA
- api | kernel32: lstrcatA
- api | kernel32: GetModuleHandleA
- api | kernel32: LoadLibraryA
- api | kernel32: EnterCriticalSection
- api | kernel32: LeaveCriticalSection
- api | kernel32: InitializeCriticalSection
- api | kernel32: GetProcessHeap
- api | kernel32: HeapFree
- api | kernel32: HeapAlloc
- api | kernel32: lstrlenA
- api | kernel32: WaitForSingleObject
- api | kernel32: ResetEvent
- api | kernel32: SetEvent
- api | kernel32: OpenEventA
- api | kernel32: CreateEventA
- api | kernel32: lstrlenA
- api | ntdll: ZwUnmapViewOfSection
- api | kernel32: IsWow64Process
- api | psapi: GetProcessImageFileNameA
- api | kernel32: CloseHandle
- api | kernel32: GetExitCodeThread
- api | kernel32: RtlMoveMemory
- api | kernel32: lstrcmpiA
- api | psapi: GetModuleBaseNameA
- api | kernel32: GetModuleHandleA
- api | kernel32: VirtualProtect
- api | kernel32: VirtualFree
- api | kernel32: VirtualAlloc
- api | kernel32: VirtualFreeEx

• Module 6 other strings

- USER32.dll
- KERNEL32.DLL
- \\9.v
- .s2'B
- OeWS
- C.xU
- n,hA5
- / AEGCuraW
- Addsss;oadLibraryA
- E`)[
- I;X2
- InitializeCr\tc
- kCount
- 0Tic
- HeapK
- *[TU

- :F8
- CH!3_o2
- SiBFObj
- dp>ewOfSec
- ZwUn6
- 6v1FD.S=
- 200 OK
- gthIo
- m-urAnczdT
- #?'1Ha?
- /a4s
- ET 3 HTTP/1.0
- Ws2_
- 32.Dl
- k>nel
- IsWow64*
- ImageFilu/<
- Opens
- veSCu7wk
- NamMl
- Enum
- cessMemyy
- EGetModuleHand
- Protec
- F"eEx
- Virtu0A
- rer.m
- bad allocation
- 9y&9
- 4kqw}
- 5;AM
- 1@22
- \\A5l
- fwixM
- U]Qd
- tH\\4f0
- IZC0J
- '8b.
- P[3e
- 8-h,[
- *a3E
- !|dCk10A
- K}9[6
-],8P
- <)6
- 8B@'
- VjD3
- o3pw
- 5\\f!
- ~1BOK
- `YWOj
- maV%
- 2jQu
- 22S2
- $$(4,
- X4iN
- t4Hu

- L7|1
- .reloc
- @.data
- `.rdata
- .text
- C.xU
- Running
- WinSock 2.0
- SI*f
- :F8
- Ws2_32
- Kernel32
- wsock32.dll
- wsock32.dll
- Content-Length:
- 200 OK
- Content-Length:
- Content-Type: application/x-www-form-urlencoded
- Host:
- HTTP/1.1
- POST
- Host:
- HTTP/1.1
- GET
- Host:
- HTTP/1.0
- GET
- bad allocation
- /arisx06.exe
- .exe
- temp
- Host:
- HTTP/1.0
- GET
- Psapi
- Ws2_32.dll
- wsock32.dll
- user32.dll
- kernel32.dll
- explorer.exe
- explorer.exe
- bad allocation
- -vej/v]7.v
- h0,@
- 50@@
- 8PQj
- D$8Pj
- Wh8B@
- 50@@
- TVjD3
- h0B@
- 50@@
- Y9|$
- 50@@
- t2S2
- h<E@
- h4E@

- h$E@
- h0D@
- SUVW
- t4Hu
- t7|1
- h4C@
- 1uKF
- 5 @@
- <)6
- 8B@'
- VjD3
- o3pw
- 5\\f!
- ~1BOK
- `YWOj
- maV%
- 2jQu
- 22S2
- $$(4,
- X4iN
- t4Hu
- L7|1
- 1uKI
- uo&v;|
- UVWhXC3
- KXq90
- UPX!
- 3.91
- `UPX2
- `UPX1
- UPX0
- C.xU
- Rich8
- !This program cannot be run in DOS mode.
- .vO!.v
- XPTPSW
- }.rd
- *HFT,
- P$oLB@
- m@l(
- c{7>cv
- &CStwtO;
- /x-w
- -Ty@
- POST #+
- d%k[
- strcmpiXRtl
- 4MGMSY_e4M
- dddd,<P\\sdddl
- V;#
- \\Z*X
- bQuqV+
- D@N`
- }fHV
- AlcSBp\tv
- m3x",?
- P'k(

- [Q<
- '#bx
- _^yT
- SSBS'
- ,PSl
- slQH
- ^G@V
- SQR
- Rh~f
- (pw`K
- hDtH
- tMhD
- H>Qx
- h@;P/
- -v!>-v
- ntdll.dll
- ntdll.dll
- HTTP/1.
- -vO!.v
- -vl"-v|
- Ht#Ht
- t%Vj
- D$ P
- D$ P
- hHB@
- D$ P
- SVWhp
- Y_^[
- h@B@
- PVVVVVV
- h(B@
- QQSVWh
- Ph~f
- XjPf
- htB@
- hlB@
- tNSj
- Rh~f
- F<_^][
- htD@
- hhD@
- hXD@
- hLD@
- tMhDD@
- = @@
- h<D@
- =(@@
- hHC@
- h@C@
- h<C@
- tjWVQP
- _^[]
- h(C@
- h|C@
- hpC@
- u\thdC@
- -$@@

- UVWhXC@
- '#bx
- _^yT
- SSBS'
- ,PSl
- slQH
- ^G@V
- SQR
- Rh~f
- (pw`K
- hDtH
- tMhD
- H>Qx
- h@;P/
- tj:QP<Z
- ,#VaGC
- h(xVM
- .\thd
- s(hmv
- FH"Q
- +'@9\ts#}
- ;JsD